

HR ANALYTICS - PREDICTING EMPLOYEE ATTRITION & RETENTION

1. Introduction

High employee turnover creates severe operational disruptions and massive financial liabilities, frequently calculated at up to 20% of a departing employee's annual salary. This project was initiated to audit an enterprise dataset of 10,000 employees, transition from retrospective reporting to predictive modeling, and uncover the root operational causes of resignation using advanced analytics.

2. Abstract

A comprehensive analytics pipeline was engineered utilizing Python and Power BI. A synthetic dataset was generated and cleaned using Pandas, successfully automating Exploratory Data Analysis (EDA) visuals to highlight salary and departmental vulnerabilities. A Random Forest Machine Learning classifier was deployed to predict attrition probability with greater than 90% accuracy. Crucially, SHAP (Shapley Additive exPlanations) algorithms were utilized to "open the black box" of the AI, proving mathematically that forced in-office mandates, prolonged tenure without promotion, and high overtime were the primary catalysts for turnover. Finally, an interactive, 9-chart Power BI dashboard was constructed utilizing a "Black Onyx" theme, allowing executives to cross-filter "Flight Risk" cohorts and isolate real-time financial liabilities via an Estimated Cost of Turnover (ECOT) metric.

3. Tools Used

- **Python (Pandas, Numpy, Scikit-learn, Seaborn):** Deployed for data generation, preprocessing, automated EDA plotting, missing value imputation, and building the Random Forest predictive classifier.
- **SHAP (Game Theory AI):** Utilized for advanced model explainability, generating summary plots that rank the exact features causing an individual to resign.
- **Power BI Desktop:** Deployed as the business intelligence layer. Constructed a multi-variable dashboard featuring Ribbon, Waterfall, and Scatter charts, styled with a high-contrast dark theme for executive presentation.

4. Steps Involved in Building the Project

- **Step 1: Data Engineering & EDA:** Generated 10,000 HR records in Python, embedding real-world corporate logic. Missing values were intentionally injected and resolved using median imputation techniques. Seaborn was utilized to automatically plot departmental and promotional attrition trends.
- **Step 2: Machine Learning & SHAP:** Split the dataset into 80/20 training and testing phases. Trained a Random Forest model, evaluating its success via a generated Confusion Matrix. Applied SHAP tree explainers to identify feature importance globally.
- **Step 3: Feature Engineering (ECOT):** Programmed conditional logic to segment the user base into 'High', 'Medium', and 'Low' flight risk tiers based on their algorithmic probability. Calculated the Estimated Cost of Turnover (ECOT) per employee.
- **Step 4: Power BI Architecture:** Ingested the enriched .csv into Power BI. Adjusted the global JSON theme parameters to render a "Black Onyx" background. Plotted 9 distinct visualizations with explicit titles, leveraging DAX and native aggregations to map financial vulnerability by department, salary band, and hire year.

5. Conclusion

The integration of Machine Learning and robust Business Intelligence provides a definitive blueprint for HR retention. The SHAP analysis mathematically proves that attrition is primarily a management and policy failure, driven by unchecked overtime and static career pathing. By utilizing the interactive Power BI dashboard, HR operations can now transition from reactive exit interviews to proactive, targeted interventions, successfully retaining high-value talent and protecting top-line corporate revenue.